

WHITECOURT, AB, Canada

WMO: 719300

Lat: 54.140N

Lon: 115.784W

Elev: 2574

StdP: 13.38

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-16.5	-28.6	1.2	-22.1	-22.4	1.8	-16.0	24.2	28.4	20.4	26.6	4.9	300	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	80.5	62.6	77.3	60.9	74.4	59.5	65.3	76.5	63.1	73.6	61.2	71.1	6.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	88.4	70.3	59.0	82.0	67.6	57.2	76.7	65.5	31.7	76.4	29.9	74.1	28.5	71.1	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	16.9	14.3	DB	-29.3	85.7	8.3	3.3	-35.2	88.1	-40.0	90.0	-44.7	91.8	-50.7	94.2	(4)
(5)			WB	-29.3	68.7	8.0	2.4	-35.0	70.4	-39.7	71.9	-44.2	73.2	-50.0	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	12.7	17.3	25.8	38.8	49.8	56.8	61.4	58.9	50.4	38.7	23.2	13.3		
	DBStd	20.44	16.36	14.01	13.21	8.85	7.78	5.52	5.36	5.81	7.55	8.14	12.02	14.07		
	HDD50	5670	1157	917	751	344	101	9	1	5	85	358	803	1139		
	HDD65	10136	1622	1337	1216	786	473	252	138	201	439	815	1253	1604		
	CDD50	1052	0	0	0	8	93	212	353	281	97	8	0	0		
	CDD65	44	0	0	0	0	1	5	25	12	1	0	0	0		
	CDH74	723	0	0	0	2	50	100	336	185	48	2	0	0		
	CDH80	107	0	0	0	0	5	11	60	26	5	0	0	0		
	WSAvg	5.8	5.8	5.7	6.1	6.6	6.7	6.1	5.8	5.1	5.3	5.7	5.3	5.6		
	PrecAvg	23.4	1.3	1.1	0.9	1.2	2.3	3.9	4.3	3.5	1.6	0.9	1.1	1.2		
(15) Precipitation	PrecMax	30.3	2.8	2.2	1.7	2.9	8.2	8.8	8.3	6.2	3.7	2.5	2.2	2.6		
	PrecMin	14.3	0.3	0.1	0.1	0.2	0.7	1.2	0.9	0.7	0.4	0.1	0.2	0.3		
	PrecStd	4.0	0.7	0.6	0.5	0.8	1.7	2.2	2.2	1.5	0.9	0.7	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	49.6	56.2	71.0	80.0	81.5	85.6	83.6	80.0	68.8	51.7	48.3		
		MCWB	39.2	39.0	42.6	50.8	56.2	61.2	65.4	64.2	61.4	53.3	42.4	38.7		
	2%	DB	42.0	43.8	50.8	63.6	73.8	76.8	81.0	78.8	73.9	60.2	45.2	40.5		
		MCWB	35.3	35.2	39.2	47.0	54.4	58.8	63.8	63.4	58.3	48.0	37.8	33.9		
	5%	DB	37.7	39.4	46.6	58.3	69.6	73.1	77.7	75.3	69.2	55.3	40.6	35.6		
		MCWB	32.8	33.0	36.8	44.1	52.3	57.3	62.4	61.6	56.3	44.6	35.2	31.3		
	10%	DB	33.5	35.3	42.7	53.7	65.6	69.7	74.4	72.0	64.1	51.0	36.8	31.6		
		MCWB	30.5	30.5	34.6	41.4	50.3	55.9	60.6	60.0	53.4	42.7	33.1	28.7		
	0.4%	WB	39.5	39.7	43.2	51.9	58.7	63.7	68.9	67.7	63.0	54.1	43.1	39.2		
		MCDB	47.2	48.1	54.6	69.2	74.4	75.3	79.8	78.4	76.4	66.8	50.4	47.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	35.7	40.0	47.7	55.8	61.0	66.1	65.0	59.5	49.2	38.8	34.5		
		MCDB	41.2	42.8	49.5	61.7	69.6	72.5	76.9	75.7	71.2	58.4	43.7	39.4		
	5%	WB	33.2	33.3	37.5	44.9	53.7	59.1	63.8	62.9	57.1	46.0	35.9	31.5		
		MCDB	37.1	39.1	45.8	57.0	67.0	69.7	74.9	72.6	67.2	53.4	39.7	35.4		
	10%	WB	30.5	30.8	35.0	42.2	51.4	57.2	61.7	60.9	54.6	43.4	33.3	28.7		
		MCDB	33.8	35.2	41.9	52.0	63.7	66.6	71.8	70.0	63.1	49.7	36.4	31.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	18.2	18.6	19.9	21.7	20.1	20.4	20.9	20.5	16.9	13.3	14.2		
		MCDBR	19.0	22.7	23.8	29.1	29.2	26.8	26.6	27.2	29.9	24.9	17.5	17.3		
		MCWBR	14.0	15.8	14.5	15.8	14.0	12.3	12.3	13.5	15.8	14.4	11.7	13.1		
		MCDBR	18.6	22.1	22.8	27.4	26.7	23.0	24.1	25.2	27.9	23.2	16.1	17.1		
	5% WB	MCWBR	14.3	15.9	14.3	15.4	13.3	11.6	12.1	13.3	15.4	14.2	11.5	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.256	0.285	0.327	0.349	0.358	0.358	0.358	0.317	0.284	0.261	0.236		
	taud		2.430	2.501	2.479	2.413	2.395	2.383	2.430	2.389	2.517	2.587	2.481	2.385		
	Ebn at Noon		240	278	289	287	285	282	280	273	273	262	232	219		
	Edh at Noon		16	20	26	32	35	36	33	33	25	19	15	13		
(44) All-Sky Solar Radiation	RadAvg		277	590	1038	1444	1728	1828	1847	1544	1074	607	292	201		
	RadStd		21	44	80	124	137	139	123	114	111	65	27	17		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page